

Technical Note: Automated CRM Client Sync

What it is

A Python pipeline that reads a daily client update CSV, validates and cleans the data, and upserts records into a local SQLite database (crm.db). A file watcher (watcher.py) triggers the sync automatically the moment a new CSV lands in the watch/ folder -- no manual steps, no scheduled polling. A Streamlit web app (app.py) provides a browser interface: upload a CSV, see the cleaned records and every data quality issue found, with no local setup required.

How it works

- Drop a CSV into watch/. The watcher detects it, runs sync.py, and moves the file to archive/ with a timestamp.
- sync.py validates column headers first (aborts if any are missing), deduplicates by client_id, cleans each field, then upserts everything in a single transaction. A mid-run crash rolls back completely.
- Three tables: clients (clean data only), sync_runs (one audit row per run), sync_issues (one row per issue -- client ID, row number, original value, resolution). Business queries never touch dirty data.

Data cleaning -- issues found and how each was handled

- **Duplicate client_id (1002 appears twice)**
Same client re-sent with an update. Kept the row with the most recent last_updated; discarded the earlier one.
- **Missing phone (1003)**
No value provided. We cannot invent a phone number we do not have. Stored NULL; logged for follow-up.
- **Comma-formatted number (1004: "2,100,000")**
Display formatting that leaked into the feed. Stripped commas; stored as numeric 2100000.
- **Wrong date format (1005: 01/06/2026)**
DD/MM/YYYY instead of YYYY-MM-DD. Parsed and normalized to 2026-06-01.
- **Invalid email (1006: mateo.garcia@example)**
Missing TLD. We cannot guess the correct domain. Stored as-is with email_valid=0; flagged for manual correction.
- **Missing portfolio value (1007)**
No value provided. We cannot invent a portfolio figure. Stored NULL; logged for follow-up.
- **Negative portfolio value (1009: -50,000)**
All other values are 640,000-2,100,000. A negative portfolio has no valid business meaning -- storing it would corrupt any average or total. NULLed out; original value preserved in sync_issues for audit.
- **Inconsistent phone format (1002: +971501234568)**
Valid UAE number, missing spaces. Normalized to +971 50 123 4568; original kept in phone_raw.

Accuracy and security over time

- Review sync_issues weekly -- recurring patterns surface provider-side problems before they compound.
- Restrict crm.db file permissions. Archive incoming CSVs after import; do not leave them in shared folders.
- If this moves to a server: credentials in environment variables, database connections over TLS.

What is most likely to go wrong

A column renamed by the provider but still present passes the header check and loads data into the wrong field silently. The fix is exact name matching rather than just presence checking. Everything else -- crashes, invalid values, duplicates -- is already handled.